

SEM	Course Code	CourseName
1	MS1990	Introduction to Materials Science and Engineering
	MS1830	Physical Metallurgy
	MS1980	Fundamentals of Scientific Computing
	EP1108	Modern Physics
	EP1118	Maths for Physics
	ME2110	Mechanics of Solids
	LA1760	Communication Skills
	MA1010	Calculus-I
	MA1020	Calculus-II
	CC1010	Introduction to Climate Change
2	MS1230	Physics of Solids
	MS1940	Materials Chemistry
	MS1054	Digital Fabrication
	CY1031	Chemistry Laboratory
	EP1211	Physics Lab
	ID1050	Artificial Intelligence
	MA1060	Elementary Linear Algebra
	MA1070	Differential Equations
	LAXxxx	Liberal Arts Elective
3	MS2330	Fundamentals of Extractive Metallurgy
	MS2120	Ironmaking
	MS2110	Semiconductor Materials
	MS2860	Computational Methods in Materials Science
	MS2970	Materials Thermodynamics and Kinetics
	MS2930	Imaging in Materials Science
	MS2280	Transport Phenomena in Materials Processing
	MS2031	Microstructure and Properties Lab (physics of solids, semiconductor materials, metallography and imaging)
	BT1010	Introduction to Life Sciences
	LAXXXX	Liberal Arts Elective
4	MS2850	Phase Transformations
	MS2870	Mechanical Behaviour of Materials
	MS2130	Steelmaking
	MS2960	Diffraction and Spectroscopy
	MS2041	Materials synthesis/extraction and characterization lab (1 Cr Ext. Met+Transport)+(1 Cr Diffraction+Spectroscopy+Mater. Synth.)
	FEXXXX	Free Elective

5	MS3XX3	Soft and Porous Materials
	MS3890	Properties of Materials
	MS3410	Electrochemical Processing of Materials
	MS3150	Corrosion
	MS3363	Powder Metallurgy Processing
	MS3041	Mechanical behavior, deformation processing and heat treatment lab (1 Cr TMP + HT) + 1 Cr Mech. Behav.)
	MSxxxx	Departmental Elective
	LAxxxx	Liberal Arts Elective
6	FExxxx	Free Elective
	MS4480	AI in Materials Science and Engineering
	MS3950	Device Fabrication and Processing
	MS3XX0	Thermomechanical Processing
	MS3315/3305/X	Internship or Project or Department Elective
	MS4481	Materials Data Science and AI Lab
	LAxxxx	Liberal Arts Elective
7	FExxxx	Free Elective
	MS4021	Functional device fabrication and electrochemical characterization lab (1 Cr Funct. Dev. Fab + 1 Cr ElectroChem/Corr.)
	MS4410	Casting and Welding
	MSxxxx	Departmental Elective
8	MSxxxx	Departmental Elective
	ID4000	Ethics and Values
	FEXXXX	Free Elective

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Segment	Credits	Basket	Sem-total
1-2	1	BE	17
3-6	2	DC	
1-4	2	BE	
	2	BS	
	2	BS	
1-6	3	BE	
	2	LA/CA	
	1	BS	
	1	BS	
	1	BE	
1-6	3	BE	15
1-4	2	BS	
1-6	2	BE	
	2	BS	
	2	BS	
	1	BE	
	1	BS	
	1	BS	
	1	LA/CA	
1-4	2	DC	19
5-6	1	DC	
3-6	2	DC	
3-6	2	DC	
1-6	3	DC	
1-4	2	DC	
3-6	2	BE	
1-6	2	DC	
5-6	1	BS	
	2	LA/CA	15
3-6	2	DC	
1-6	3	DC	
1-4	2	DC	
1-4	2	DC	
1-6	2	DC	
	4	FE	

1-6	3	DC	19
3-6	2	DC	
1-4	2	DC	
5-6	1	DC	
1-2	1	DC	
1-6	2	DC	
1-6	3	DE	
	2	LA/CA	21
	3	FE	
1-6	3	DC	
1-6	3	DC	
1-4	2	DC	
1-6	6	DE	
4-6	1	DC	
	2	LA/CA	13
	4	FE	
1-6	2	DC	
1-6	3	DC	
1-6	8	DE	10
1-6	6	DE	
	1	LA/CA	
1-6	3	FE	
		Total	129